DEVELOPMENT OF A MULTI-FACET FACIAL RECOGNITION MODEL FOR INSTITUTIONAL ATTENDANCE ACTIVITIES

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ABSTRACT

Conventional techniques for tracking student attendance in educational institutions typically rely on the old-fashioned manual attendance method, the use of physical sign-in sheets, or simple biometrics such as fingerprint scanning. The methods employed in traditional attendance monitoring are known to be laborious, disruptive to lessons, vulnerable to human fraud, and unhygienic in the case of a large group of students. New developments in contactless technologies have made it possible to use QR codes with web-based applications but the novel method still has its issues including the need to avoid queues, the chance of fraud detection, and the requirement for someone to point a QR code this person's phone.

The Multi-Facet Institutional Attendance System (MFIAS) is being introduced in this paper as a contactless attendance system that is dependent on schedule which works all problems present in conventional systems. The manual scanning of people during attendance is not applicable any more thanks to the Automatic Departmental Timetable Engine which is scheduling classes and monitors them according to it. Final identity verification is performed by means of the multi-faceted computer vision that considers many real world possible hindrances like dynamic lighting in classrooms, various head position angles (tilt ± 15 degrees), and many more things that may block the identification (like eyeglasses and beard). The process employs the usage of Haar Cascade framework for the image detection process, Dlib 68 points model for landmarks issues, Residual Neural Network RegNet-34 for defining facial features, and KNN (k=3) method.

Additionally, it contains asynchronous multi-student batch checking, precise date and time recording for arrival and departure as well as duration calculating. The attendance management software features strict authorization at the administrator level which makes it possible to update attendance record only by the appropriate people. Extensive experiments show that the video classification from still pictures as well as from the live and previously recorded video gives the accuracy of 99.29 percent, 0 percent false omission rate and 6 seconds average verification time despite the changes in videos. Also, the cloud attendance software is able to produce tamper-proof log files and convert them into CSV, Excel and PDF format files that helps assure the safety of attendance taking process through biometrics.

I. INTRODUCTION

Attendance records in schools are considered to be vital part of education management. Such records are known for helping in various internal evaluations, approval to conduct research in laboratories, and accreditation of examinations in educational establishments. Nonetheless despite numerous innovations in modern-day technologies participation is rather old-fashioned and is taken mainly in shape of roll call when people are asked to fill out sheets with signatures or go through the process of using gadgets for collection of information about attendance such as fingerprint scanners. Despite the high

popularity of the methods that are used in this area they cannot be called efficient because roll calls may take up to 10-18 percent of class time; paper sheets are still subject to the dangers of buddy signing, and gadgets require physical contact and cannot be used in all situations.

To overcome these challenges, creative approaches were introduced in the form of mobile applications and QR codes scanning systems. In the traditional QR code approach, a teacher generates a code which has a period after which it expires and a student scans it to show that he is present. However, research shows that QR codes did not eliminate cheating, they just made it different since students can taking pictures of a QR code on the screen and send it to their friends outside the university. Also, the necessity for students to locate and open the application, log in to the network using the application and point the camera at the screen introduces additional difficulties and delays into the process of teaching.

Biometric facial recognition technology represents the primary way of performing academic identity verification without direct contact. After capturing human facial features, they cannot go missing or be altered; therefore, biometric facial recognition technology allows automatically keeping track of students in class just by watching them coming into the classroom. However, ready-to-use face recognition systems perform significantly worse in free unstructured educational environments. In ordinary classrooms students' faces may present a great variety of recognizable and unrecognizable features.

- **Changing Brightness:** Daylight differs from shining sunlight from the outside to dim light inside classrooms.
- **Tilting in Movement:** Students do not walk straight when passing through the doors of classrooms.
- **Facial Overlapping:** Common accessories of people, for example, eyeglasses with heavy glass, different types of beards and hats cover some important geometric reference points.
- **Delay in the Process of Ingress:** The usage of complex end-to-end implementation models causes significant costs resulting in the delay of data ingressing into the classroom.

This study develops an automated, contactless Multi-Facet Facial Recognition Attendance System to eliminate the challenges associated with existing limitations without requiring students to scan QR codes. The system comprises five important parts:

1. The Timetable-Based Punctuality System allows direct access to the department's timetable which allows to automatically plan attendance sessions without relying on QR codes.

2. Advanced Biometric Algorithms allow creating sophisticated data processing systems involving Haar Cascade classifiers, Dlib's 68-point face modeling method, ResNet-34, and K-Nearest Neighbor algorithms.

3. The usage of Simultaneous Student Verification allows the acquisition of information from the video stream and recording of the concurrent presence of learners.

4. Dynamic Audit of Entry and Exit Times enables registration of the time of arrival and departure of students from the classroom, creating the complete record of their presence in class.

5. Admin Security prevents anyone else from performing writing and reporting operations within the database.

The rest of the paper is organized into sections. The following section describes the previously conducted research and the methods which form the basis for the technique described in this paper. This section is anything but a general outline of the mathematical approaches and methods utilized in the realization of the project. The section IV gives the evaluation of the technique and includes the practical assessment of the technology used. The next section V contains the evaluation of the technical part of the system as well as organizational issues. Finally, section VI gives the conclusions and recommendations for the future research.

II. RELATED WORK & RESEARCH FOUNDATIONS

Over the last 20 years, automated attendance systems have come a long way. In the early days, these systems made use of the Viola-Jones object detection algorithm together with either Local Binary Pattern Histograms (LBPH) or Principal Component Analysis (PCA) to overcome the challenges related to capturing localized facial features. While these systems do not require heavy computational resources and allow the implementation of attendance systems on microcontrollers, they fail in extreme lighting situations, slight head movements and obscured faces.

Automated attendance systems have made great progress in the last 20 years. At first, this technology was based on Viola-Jones object detection algorithm combined with Local Binary Pattern Histograms (LBPH) and Principal Component Analysis (PCA). Hence, the algorithms did not need much power and, therefore, were usable even on microcontrollers, but they still did not manage to produce reliable results in difficult lighting conditions, in particular, when head movements or obscured faces were present.

The emergence of deep convolutional neural networks (CNNs) enabled researchers to switch to high-dimensional representation methods of analysis. Pattnaik and Mohanty conducted an evaluation of cloud-based computer vision services such as Google Cloud Vision or Amazon Rekognition that proved to have high accuracy in picture analysis. When switching to real-time video transmission the disadvantages of such methods became obvious: video-steaming creates heavy traffic load and also has untrusted latency. Moreover, in the case, where privacy is an issue, it requires stable connection to the Internet of the university's camera. According to the study performed by Lateef and Kamil, it became possible to track many people's faces in one smart classroom with the help of different YOLOv7 models.

Nevertheless, the fact that depth detection devices must operate at real-time frame rates necessitates the need for expensive GPU technology, making these products impractical for use in schools, as well. So as to cope with a certain kind of blocking of the face, Kamil and his co-authors came up with a system for online authentication that is capable of detecting individuals that have protective masks as required by the condition of the pandemic, achieving an efficiency of 81.8% in identification of such individuals. Nevertheless, this system is not good for solving the case of complicated distortions and is aimed only at solving the problem of identifying the lower part of the face with the nose covered with a mask. In addition, Afiyanto and his co-authors substantiated the effect of variation of illumination on the metric Euclidean distance which is reduced by 34% within the classrooms that are constantly illuminated by 150 lux and less.

The research by Jack and others into the field of multi-dimensional analysis of attendance management has resulted in a major breakthrough that solves many existing problems. According to the claims of Jack et al., the decoupled hybrid solution composed of Haar Cascade detection, Dlib landmark alignment, ResNet-34 128-dimensional vector encoding and distance-weighted KNN classifier shows 99.29% recognition rate in video streams and 6.00 seconds average verification time, while the other deep learning models are encountered with the obstacles caused by the shortcomings of the inference process.

TABLE I. Comparative Summary of Automated Attendance Literature

Reference	Core Methodology	Primary Advantage	Identified Limitations & Gaps
Bairagi et al. (2021)	Haar + LBPH	Minimal cost; fast writes.	Fails under off-pose & lighting shifts.
Trivedi et al. (2022)	OpenCV + Eigenfaces	Automates SQL logging.	Rigid alignment; zero occlusion tolerance.
Kamil et al. (2023)	SVM + Mask CNN	Masked face verification.	Isolated occlusions only; no pose handling.
Lateef & Kamil (2023)	YOLOv7 Deep Model	Continuous multi-face tracking.	Heavy GPU load; high latency at edge nodes.
Afiyanto et al. (2025)	Dynamic Filter + CNN	Illumination normalization.	High overhead; fails on composite angles.
Jack et al. (2026)	Haar + ResNet-128D + KNN	Multi-facet robustness (99.29%).	6.00s latency; lacks timetable automation.
Proposed MFIAS	**Timetable + ResNet + KNN**	**Zero QR; batching; In/Out logs.**	**Needs calibrated entry camera placement.**

III. PROPOSED SYSTEM ARCHITECTURE & METHODOLOGY

The Multi-Facet Institutional Attendance System or MFIAS has been developed with six modules that are interrelated to each other in a unique way. The modules are; (i) Student Biometric Enrollment Layer; (ii) Departmental Timetable Scheduling Engine; (iii) Multi-Students Computer Vision Ingestion Layer; (iv) Multi-Facet Deep Feature Embedding and Classification Engine; (v) In/Out Temporal Tracking and State Machine; and (vi) Role-Based Admin Governance and Multi-Format Export Module.

A.Student Enrollment & Biometric Blueprint Synthesis

Enrollment can be viewed as an administrative procedure that takes place before the operation of classes begins. Every student accepted by the institution is registered with his/her demographic details such as Roll Number, Name, Department, Year, and Academic Section. Enrollment is carried out in front of the camera, which captures 30 different images of the student's face.Frontal gaze under ambient and supplemented lighting.

- Yaw motion involves tilting forward and backward (by 15°).

• The study would consider the impact of wearing and not wearing glasses along with different types of facial hair.

The images indexed are processed through bilinear interpolation to achieve a regulated affine alignment of 400 × 400 pixels in size. All images are then converted to single-channel 8-bit grayscale format, thus effectively capturing the texture and geometry while disregarding color variations. Then, facial landmark points are extracted, converted to 128-dimensional embedding vectors by the ResNet-34 deep learning backbone, and stored in an indexed binary file through Pickle available in the Python environment. This way, the need for deep learning training in real time has been eliminated due to the pre-computed embeddings.

B. Automated Departmental Timetable Engine

A significant advancement brought about by MFIAS is the removal of the reliance on QR codes and mobile scans inside the elements. The attendance of students should adhere to the established formal schedule of classes. The Timetable Engine functions as an independent daemon thread interacting with the university database. The engine relies on the master schema of relationships:

```
TIMETABLE_SLOT = {
"slot_id": UUID,
"department": "AI&DS",
"semester": 6,
"section": "A",
"day_of_week": "MONDAY",
"start_time": "09:00:00",
"end_time": "09:50:00",
"grace_period_mins": 10,
"course_code": "AD8601",
"course_title": "Computer Vision & Deep Learning",
"assigned_faculty_id": "FAC_9519_01"
}
```

The hardware clock is compared to the timetable index all the time:

1. Pre-Session Trigger: In advance of the time of the session by 2 minutes, the system starts the camera pipeline, sets buffer streams, and saves the biometric vectors of students who are enrolled in the class from the respective department and semester.

2. Effective Presence Window: During the period from initial time to final time, video feeds are analyzed to assess student presence. Students verified during grace minutes are marked as Present; verifications occurring after the grace time are designated Late.

3. Completion of Session: After end time, the camera feeds are automatically disabled. Any student participating in the group who was absent during that period is logged as Absent. The final attendance log is saved.

C. Multiperson Face Recognition and Alignment

For the whole time of the session, the camera output is recorded at a rate of 30 frames per second (fps). Unlike the single portal cases, it should be noted that in reality the students do not walk alone. The platform uses OpenCV pre-trained frontal Haar Cascade classifier for the processing of the video stream. The system uses integral image representation to assess rectangular Haar-like features through a multi-scale sliding window:

$$ii(x, y) = \sum_{x' \le x,\, y' \le y} i(x', y') \quad (1)$$

In this notation, ii(x,y) refers to the cumulative value associated with facial recognition, while i(x',y') indicates the pixel intensity. The methodology is based on a series of AdaBoost classifiers eliminating undesirable image backgrounds in the earlier stages of analysis. All the bounding boxes Bk = [xk, yk, wk, hk] are acquired at once.

To achieve the required consistency despite head inclinations up to ±15°, the bounding box coordinates are aligned to the facial features of the person. A Dlib ensemble of regressor decision trees is used to determine the location of 68 key facial features including chin, eyebrows, nose, eyes, and mouth. The transformation between the inner corners of the eyes (features 37 and 46) and the tip of the nose (feature 31) is calculated and is used to transform the face to a frontal position.

D.Deep Feature Extraction & ResNet-34 128-D Metric Encoding

Affinities of normalized facial regions are learned using the Residual Neural Network (ResNet-34) architecture. Instead of performing identity classification like a softmax cross-entropy classification problem, our idea is to use the network purely as a feature extractor and convert facial landmarks based on geometry into 128-dimensional Euclidean space:

$$f: I \to \Re^{128} \quad ext\{such\ that\} \quad \|f(I)\|_2 = 1 \quad (2)$$

The ResNet-34 architecture has shortcut identity connections over 34 convolutional layers, allowing us to train deep networks without the problem of gradient vanishing. The weights are trained with the aid of the Triplet Loss method. The triplet Loss function makes it necessary for similar facial representations (anchor f(xia) and positive case f(xip) to be grouped close to each other and meaning that different facial representations (negative case f(xin)) must be separated from each other with a predetermined distance α:

$$L = \sum_i^N max(0, \|f(x_i^a) - f(x_i^p)\|_2^2 - \|f(x_i^a) - f(x_i^n)\|_2^2 + \alpha) \quad (3)$$

In our case, the value of α is equal to 0.2. By using this approach, the variations of the similar person facial appearance caused by eyeglasses, beard, or bad lighting create a minimal shift in the vector that is still located within the same identity area but separate from the identities of

other students because of the high distance in Euclidean space.

E. Distance-Weighted K-Nearest Neighbors (KNN) & Unknown Rejection

For live identity assignment, the normalized 128-dimensional vector x undergoes classification against enrollment symbols recorded in the database using the distance-weighted K-Nearest Neighbors algorithm with k = 3. The Euclidean distance calculated between this test vector and the training vector xj can be written as:

$$d(x, x_j) = \sqrt{(\sum_{m=1}^{128} (x_m - x_{j,m})^2)}$$

In order to minimize the impact of outliers caused by changing lighting conditions, the voting weights wj are assigned inversely proportional to the square of the Euclidean distance to the enrolled individual:

$$w_j = 1 / (d(x, x_j)^2 + \varepsilon)$$

Where ε = 10-5 avoids division by zero. The predicted identity y is found through the majority voting mechanism:

$$y = argmax_c \sum_{j=1}^{k} I(y_j = c) \cdot w_j$$

The Unknown Person Protocol: In order to avoid the cases of false positive identification, strict limitations regarding the Euclidean distance threshold are established.

```
IF min(d(x, x_j)) <= 0.60:
    Identity = y (Valid Student
Verified)
ELSE:
    Identity = "UNKNOWN / INVALID
STUDENT"
    Trigger Alert Flag & Log
Unregistered Face Capture
```

Temporal In/Out State Machine & Session Tracking

Contrary to single-instance check-ins, the requirement of educational accountability entails checking the time spent by each student inside the classroom. The MFIAS database captures 2 separate time stamps for individual students for each session.

1.In-Time (Entry Timestamp): This time stamp is generated as soon as a student is detected in the ongoing session for the first time:

entry_timestamp = CURRENT_TIMESTAMP.

2. Out-Time (Exit Timestamp): The last time stamp is updated every time a student is detected in the last 10 minutes of the session or when he/she leaves the class: exit_timestamp = CURRENT_TIMESTAMP.

3. Duration Calculation: Once the session is over, the duration is calculated as follows: duration_minutes = (exit_timestamp - entry_timestamp)/60

4. Attendance Status:

○ If entry_timestamp ≤ (start_time + grace_period) and duration_minutes ≥ 0.75 × session length, then it is recorded as PRESENT.

○ If entry_timestamp > (start_time + grace_period) and duration_minutes ≥ 0.75 × session length, then it is recorded as LATE.

○ If it is neither detected nor duration_minutes < 0.50 × session length it is recorded as ABSENT.

G. Role-Based Administrative Governance & Multi-Format Reports

It is important for the security integrity to ensure that neither faculty members nor students can modify attendance records. MFIAS also uses JWT authentication-based RBAC.

● Student/Public Role – Read-only access to personal attendance percentages.

● Faculty role – Read-only access to session summaries and attendance records in the classroom.

● Admin role – Only one who has the ability to reset attendance flags (such as being on duty). Administrative changes are recorded through explanation codes whereby the ID of the administrator is also logged in the system.

For archiving purposes, the website has embedded report generators that automatically generate reports in three different formats.

● CSV – Spreadsheet format with raw data.

● XLSX – Tables with colors coded status stamps and statistics of attendance.

● PDF – Printed reports with signatures.

IV. Experimental Setup, Results & Performance Evaluation

A.Experimental Environment & Dataset Curation

Real-life testing in an academic environment was simulated in a laboratory setting. Technology used in the computational testbed included an Intel Core i7 CPU, 16GB DDR4 RAM, and NVIDIA GeForce RTX 3060 GPU (for the ResNet embedding) and the processing of the CPU was employed for answering the question of whether edge computing is feasible.

The experimental data were collected utilizing the method presented by Jack et al. (2026). The images were created with various experimental conditions in mind (illumination (bright vs dim), head position (facing forward, tilted to the right or left at a 15-degree angle), clothing (with

or without glasses and beard). The evaluation included more than 3000 still images, over 5173 video frames, and real-life tests of continuous student entry.

B. Detection Stage Evaluation: Failure to Detect (FTD) Dynamics

System evaluation requires decoupling face detection from classification. Table II outlines the detection efficacy of the pre-trained Haar Cascade classifier across static frames and continuous video.

TABLE II. Haar Cascade Face Detection Performance Metrics

Evaluation Modality	Total Samples	Detected Samples	Detection Rate (%)	FTD Rate (%)
Static Image Dataset	3,000	1,750	58.33%	41.67%
Continuous Video Frames	5,173	4,059	78.46%	21.54%

It was noted that static photographs had an FTD of 41.67%, partly as a result of acute head positions and dim areas of low contrast that did exceed gradient limits set by Haar-like filters. In contrast, during regular video transmission (30 fps), the FTD dropped to 21.54%. Importantly, an FTD event does not result in a false acceptance; it simply rejects a frame that is unsuitable for further processing. Due to the temporally redundant nature of live broadcasts, a passerby walking through a doorway will yield anywhere from 60 to 120 frames for accurate detection as the student adopts a natural position.

C. Evaluation of Multilevel Recognition Performance

To evaluate the quality of classification, 4,059 frames correctly localized over video recordings were examined. The particulars of precision, recall, and F1-score for each student group are presented in Table III.

TABLE III. Recognition Performance Metrics on Video Frames

Student Identifier	Frames	Precision	Recall	F1-Score	Accuracy (%)
Student Cohort 1	1,215	0.9959	0.9934	0.9946	99.41%
Student Cohort 2	492	0.9979	0.9593	0.9782	98.78%
Student Cohort 3	2,352	0.9903	0.9996	0.9949	99.36%
Cumulative Average	**4,059**	**0.9947**	**0.9841**	**0.9892**	**99.29%**

The system demonstrated that the 128-dimensional ResNet-34 embeddings yield good separation between different classes while remaining close within classes regardless of the type of accessory or slight motion blur, since the overall recognition accuracy achieved by the system was 99.29%.

A. Classifier Testing: Comparison of KNN and Other Machine Learning Models To confirm the appropriateness of using distance-weighted KNN as the architecture, the same 128-dimensional embedding vectors were used to assess the classification accuracy of a Linear Support Vector Machine (SVM) and a Shallow Multi-Layer Perceptron (MLP with a hidden layer of 64 neurons running in Adam optimization mode).

TABLE IV. Comparative Benchmarking of Lightweight Classifiers

Classifier Head	Static Acc (%)	Video Acc (%)	Precision	Recall	F1-Score	Retraining Need
Linear SVM	99.94%	99.12%	0.9937	0.9749	0.9842	Full Retrain
Shallow MLP (64)	99.66%	99.25%	0.9937	0.9812	0.9931	Iterative BP
Weighted KNN (Ours)	**99.94%**	**99.29%**	**0.9947**	**0.9841**	**0.9892**	**Instant Insert**

Despite their accuracy scores being only slightly below that of KNN (99.12% and 99.25% for SVM and MLP respectively), in practice, the algorithms of both of these models should be re-tuned across the entire student base whenever a new student appears. In this regard, the distance-weighted KNN is a non-parametric algorithm that allows for an unrestricted addition of new students into the search space without affecting the speed of inference. *Response Latency & Multi-Facet Operational Robustness The real-time working capacity of the system was analyzed in different conditions to determine its response speed and False Omission Rates (FOR). The data is provided in Table V.*

A.TABLE V. Average Response Time (Seconds) Under Multi-Facet Variance

Illumination Setting	Head Orientation	With Glasses / Beard	Without Glasses / Beard	Avg Latency (s)
Bright (>450 lux)	Frontal Straight (0°)	3.79 s	3.81 s	3.80 s
Bright (>450 lux)	Yaw Tilt (±15°)	4.37 s	4.11 s	4.24 s
Dim (<150 lux)	Frontal Straight (0°)	7.84 s	4.37 s	6.10 s
Dim (<150 lux)	Yaw Tilt (±15°)	14.40 s	13.15 s	13.77 s
Composite System Mean Response Time Across All Operational Scenarios				**6.00 s**

Given that the identity check is executed under very bright light, the execution lasts only 3.80 seconds. However, in the conditions of low light and yaw rotation, the execution time grows to 13.77 seconds due to localized searches of features in low-gradient pixels. In general, the system performance in all conditions was about 6.00 seconds, which is very important since the FOR was 0% meaning that each student passing the monitored checkpoint was identified properly.

B. Simultaneous Multi-Student Batch Verification In this part of the research, multi-student batches of 2, 4, and 6 students during the passage of the checkpoint were analyzed in terms of effectiveness. The multithreaded architecture allows the system to crop the bounding box and make the feature projection via ResNet at the same time. The results can be seen in Table-VI.

TABLE VI. Simultaneous Multi-Student Verification Throughput

Simultaneous Faces	Batch Detect (ms)	Parallel 128-D (ms)	KNN Match (ms)	Total Cycle (ms)	Batch Accuracy (%)
1 Student	42 ms	118 ms	4 ms	164 ms	100.0%
2 Students	58 ms	234 ms	7 ms	299 ms	100.0%
4 Students	84 ms	462 ms	14 ms	560 ms	98.5%
6 Students	112 ms	688 ms	22 ms	822 ms	97.2%

V.DISCUSSION & PRACTICAL INSTITUTIONAL IMPLEMENTATION

The empirical evidence shows that the Multi-Facet Institutional Attendance System (MFIAS) solves effectively the usability and security issues that have faced earlier attendance systems.

- The problem of QR Code Vulnerabilities has been solved. Unlike conventional QR-based systems necessitating manual student initiation with the aid of shared pictures, in MFIAS an autonomously acting Timetable Engine activates each session, thus eliminating the danger of proxy attendance.
- Resilience under Multi-Factor Variations. By synthesizing Dlib 68-landmark structural positioning with ResNet-34 metric embedding, MFIAS guarantees 99.29% recognition accuracy despite various disturbances introduced by the students daily, including glasses wearing, growing beard, and head movements.
- There are no false omissions even when MFIAS works continuously. Despite the fact that Haar Cascade detection may display some static frame misses if applied at intricate angles, the timely redundancy of continuous 30 fps video stream allows authenticating all legitimate students during a 6 second period with zero false omissions.
- Data Governance and Auditability: In order to prevent any manipulation of grades, writing access is limited to the administrators alone which ensures that unauthorized changes cannot be made. At the same time, rapid reporting of information across multiple vectors, including CSV, Excel, and PDF meets the regulatory requirements of the universities.
- Practical Deployment Considerations: When deploying MFIAS technology in large lecture halls hosting more than 200 students, using a single camera system may lead to delays during the five-minute peak hour traffic. In support of this view, the option to have a second camera installed alongside the current setup will enable the possibility to ensure faster data processing through having multiple access points.

VI.CONCLUSION & FUTURE SCOPE

The article outlines the architecture, implementation, and empirical assessment of the MFIAS, a self-reliant and non-contact facial recognition technology. The introduction of the Automated Department Timetable Engine has eliminated manual methods of QR scanning in the educational institutions environment. At the core of the system is biometrics technology, utilizing Haar Cascade face detection, Dlib 68 facial points alignment, ResNet-34 128-dimensional vector formation, and K-NN Distance weights classification. The use of this system provides an amazing 99.29% accuracy in video recognition, 6 seconds response time, and 0% omissions despite various lighting conditions, poses, and accessories.

Furthermore, the system tracks automated In/Out times with auditable administrative access via reports in various formats.

Future enhancements plan to expand the developments in three areas:

(i) lightweight infrared (IR) depth camera technology with eye blink CNN classifiers for hardware-level anti-spoofing;

(ii) (ii) installation of multi-camera edge nodes with TensorRT acceleration for reducing inference latency to under one second in big auditoriums; and (iii) incorporation of predictive analytics technology for the early identification of cases of chronic student absence.

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